

MAHMUD ELGOUERI

Systems & Hybrid Infrastructure Engineer

Tripoli, Libya • +218 92 778 5022 • mahmudeg2000@gmail.com

linkedin.com/in/mahmud-elgoueri • github.com/mahmudeg • mahmudeg.github.io

PROFESSIONAL SUMMARY

Systems and hybrid infrastructure engineer with a Telecommunications Engineering background and a year of enterprise delivery in the oil & gas sector. I design and support hybrid environments spanning on-premises Windows Server and Microsoft Azure / Entra ID, harden Active Directory and Group Policy for security compliance, and deliver network segmentation across switches and firewalls. Equally comfortable in Linux, Docker, and monitoring pipelines (Prometheus, Grafana). Experienced in customer-facing deployment and knowledge transfer to client IT teams, and in producing the design documentation and runbooks behind each rollout. Microsoft, Fortinet, and Huawei certified.

Technical Skills

Identity & Directory: Active Directory DS, Group Policy design & auditing, hybrid identity, Entra Connect / Cloud Sync, Microsoft Entra ID

Cloud & Hybrid: Microsoft Azure (AZ-104), hybrid cloud / hybrid data center, Azure-on-prem synchronization, Microsoft 365

Windows & Virtualization: Windows Server 2016–2025, VMware / vCenter, PowerShell automation

Networking & Security: VLAN design & segmentation, FortiGate, pfSense, policy-based firewall rules, security hardening, network setup

Linux, Containers & Data: Ubuntu / Linux administration, Docker & Docker Compose, Prometheus, Grafana, shell scripting

Tools & Delivery: Visio, Word, Excel, PowerPoint; design documentation, runbooks, technical assessments, knowledge transfer

Languages: Arabic (native), English (professional working proficiency)

PROFESSIONAL EXPERIENCE

Systems & Hybrid Infrastructure Engineer — Electron Technology Solutions

Jun 2025 – Present

Tripoli, Libya | Client sector: Oil & Gas

- Delivered a new enterprise Active Directory domain environment across 6 geographically distributed field sites for a national oil & gas operator; travelled on-site to 3 fields to deploy the solution and provide hands-on instruction to the local IT teams.
- Designed and audited Group Policy Objects to enforce security compliance and standardise configuration baselines across the enterprise, reducing configuration drift between sites.
- Managed synchronisation of on-premises identities and resources with Microsoft Azure / Entra ID, maintaining a hybrid identity model with consistent access to resources across on-prem and cloud.
- Assessed existing client environments and issued improvement recommendations covering identity, configuration, and security posture.
- Delivered a network segmentation project on a live production network with minimum downtime — isolating VLANs and enforcing policy-based connectivity across switches and FortiGate firewalls.
- Authored and maintained design documents, deployment runbooks, and handover material enabling client teams to operate and extend the environment independently.

KEY PROJECTS

5G Standalone Core — Containerised Deployment & KPI Monitoring Pipeline

2025

Graduation project, University of Tripoli

- Deployed a full 5G standalone core as a multi-container application on Ubuntu using Docker, defining the service topology, networking, and configuration for each network function.
- Instrumented the deployment to collect and store time-series metrics in Prometheus, then transformed the data into KPI dashboards in Grafana for performance analysis and monitoring.
- Validated the deployment against functional and performance requirements and documented the architecture, configuration, and results.

Enterprise Domain Rollout Across Multiple Field Sites

2025

Professional project, oil & gas client

- Standardised identity, Group Policy, and security configuration across 6 sites; produced hardening checklists and site-onboarding documentation for repeatable deployment.

Production Network Segmentation

2025

Professional project, client production network

- Isolated VLANs and enforced policy-based firewall connectivity on a live network with minimal downtime, using switch and FortiGate configuration.

Home Lab & Technical Blog

Ongoing

mahmudeg.github.io | github.com/mahmudeg

- Maintain a virtualisation and Docker lab and publish technical write-ups on infrastructure, cloud, and security; currently extending into SQL, data pipelines, and Kubernetes.

EDUCATION

B.Sc. Electrical & Electronic Engineering (Telecommunications)

Spring 2025

University of Tripoli — Tripoli, Libya

CERTIFICATIONS

- Microsoft Certified: Windows Server Hybrid Administrator Associate (AZ-800 & AZ-801) — 2026
- Microsoft Certified: Azure Administrator Associate (AZ-104) — 2026
- HCIA-Security — Huawei, 2025
- Fortinet Certified Associate (FCA) in Cybersecurity — FortiGate 7.4 Operator
- Cybersecurity Boot Camp — LATI & NETSCOUT (6-week intensive, hands-on)